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Teaching Programming by Generating Business Activity

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Abstract

Today's economic slowdown demands pedagogical skills to be business more friendly. One way to achieve this need is to integrate computing skills taught to the students with the development of commercial products. In this course of computer graphics, an attempt has been made to teach Visual C++ (VC) by supplementing the lab activities using a novel "code conversion" approach and by engaging the students in the activities of a human artist. Students develop and modify the previously worked out Turbo C (TC) projects of 2D landscapes into VC. This work mainly emphasizes illustrating the dual role of these projects in boosting the learning of the programming language and in generating a business activity through an exhibition organized in a period of 10 weeks. Results show that all student groups were able to complete their projects with a competitive fervor**.** A comparative study also has been carried out.

1. Introduction

World economies are under pressure due to the global recession. This situation has caused financial crises and led to tremendous job losses. Should computer teachers play a role in the economic revival process? How this can be achieved in a semester-based programming course without compromising the teaching/learning process? As wars cannot be won only by the efforts of an army, the entire society owes participation in such circumstances. In the same spirit, in this bleak economic period salvaging cannot be left alone to the financial institutions. Teachers must step forward to assist with this crisis.

Based upon this idea, the author taught a computer graphics course in fall 2009. The objective was to teach the students and then give them an opportunity to engage in a business activity via commercializing their work. Their work should result in an innovative, cost effective, programming based, and graphic oriented product. To achieve these tetrahedral characteristics, the computer artist domain seemed a good choice.

At Sir Syed University, the computer graphics (CG) course is taught in the third semester of computer engineering department and consists of two labs and two theory hours per week. During labs, normally Turbo C (TC) is used during first eight weeks and 3dMax in the remaining eight weeks. 3dMax is fine but the TC programming language has almost become obsolete in the software industry. Earlier in 2008, the author taught this course and students were asked to develop landscapes depicting the theme of "Sunset Beauty" using TC as the final project. All these projects were archived. This time it was decided to teach this course in a business context. Thus, a two-prong strategy was chalked out: one for generating the business activity and the other to satisfy the academic requirements. The former is achieved by informing the students that their computer art would be displayed in an exhibition where people can buy their work. The later was accomplished by asking the students to transform their TC projects into Visual C++ (VC) using a technique known as code conversion technique (CCT). In this regard students were required to maintain a CCT document which is discussed later in the methodology section. Students were required to complete their VC projects in a limited time of eight weeks and the exhibition was set to be organized in the 10th week.

This is a descriptive research and the population includes CG students taught by the author during the years 2008-2009. Apart from economic gains, the author believes that the technique used in this research can help students in learning a difficult programming language like VC. Results are based upon a group based project performance, CCT document, scope of the economic activity, revenue generation, impact of exhibition, and students' and visitors' comments. This paper describes the background work, research methodology, sample lab activities coalesced with code conversion assignments, exhibition details, the associated business activity, and the results of this study.

2. Background Work

Using VC for teaching computer graphics is a difficult proposition. VC is famous for its complex and huge coding library (MFC). No prior knowledge of GUI, event-driven programming, and OOP make the situation even more difficult. To deal with this situation, it was decided to omit the above mentioned VC programming concepts (Gibbons, 2002). This decision undoubtedly simplified the path for achieving the goal of the course. It is worth mentioning that in this research VC is taught in the third semester. However, this language and other OOP variants like C# and Java are normally taught in the third and final year courses of the BS program (Athauda, et al., 2005). This justifies the omission strategy being pursued in this course. Various noteworthy steps have been taken in previous research for teaching programming. However, no effort has yet been made to give an exposure to the students for organizing an exhibition in order to sell their work. In a programming course, students develop a lot of software projects but often these projects go to waste as no effort is done to use them afterwards. In this context, Khan (2005) has shown how student projects can be transformed into a commercially viable product using the concept of stepwise refinement. Neuhaus (2008) related about computer graphics, in order to promote business activity, students are assigned community projects. The novelty of this research lies in the fact that students are asked to develop a class project using their imagination and creativity. Then a forum is provided for generating the business activity. The project development process incorporates the CCT. Surveying literature does not report about the adoption of this technique for teaching programming and is normally used by software agents and case tools for code migration. Orientation towards the business activity in this study unintentionally introduced an entrepreneurial behavior. Student projects in this context can be treated as entrepreneurial ventures. Recently entrepreneurial education has received great prominence in many universities (Hargadon, 2010) due economic reasons. The teaching style of courses in this discipline deviates from the traditional lecture based approach (Sogunro et al., 2004). Lope and Abdullah (2009) have conducted a study to explore the important teaching techniques to improve entrepreneurial skills. According to their study, the top ranked teaching technique is running a real business. Feathermore (2007) argued that entrepreneurship being an innate quality of a person cannot be taught. Good teachers can only harvest an entrepreneurial spark among their students. This aspect has been demonstrated unexpectedly in this research.

3. Methodology

The methodology adopted for teaching the course has business as well as technical aspects. The business appetite is fulfilled by holding an exhibition while the technical concerns are met through the CCT. This approach is in contrast to what was done in teaching the same course in 2008. Again the students were given separate projects but the objective is to add more complexity to this project and to employ it for incorporating business flare in the course. Important ingredients of the methodology are discussed below.

3.1 Group Formation

Consistent with the CG taught in 2008, the instructor divided the students into non-homogenous groups (Faria, 2006) of four each. Twenty-five groups were formed. Absent students, who often indulge in parasitic attitude (Kouznetsova, 2009) were kept together, resulting in two additional groups. However, these two groups are not part of the sample due to obvious reasons.

3.2 CCT

CCT was the technique adopted to transform TC projects into VC. To incorporate CCT in each lab, students were provided a brief document containing the list of at least three Turbo C commands. Students were then asked to search their VC counterparts using MSDN (Microsoft Development Network). After eight weeks, each student had this document checked and marked at least twice, first in the forward sequencing order and then in the reverse sequencing order of their roll numbers. The two checking orders are designated as Round1 and Round2. The checking process is not time consuming considering the material, time span of the lab, availability and involvement of two faculty members for the whole eight weeks. This document is required to specify the VC command name (i.e., the method name), its parameter list, class name, and its invocation using the dot operator. Only one mark is allocated for each round so that student work is not influenced by marks.

3.3 Projects

Again the final semester project is similar to what was done previously in 2008. The difference lies in using VC by implementing CCT. Also, it was made mandatory that each group has to inject some new real world objects in their work, which was not part of the original TC projects assigned to them.

3.4 Computer Artist Exhibition

The guidelines for holding the exhibition included that it should not disrupt the normal classes. Indeed, it is a tough task to hold such an event in the middle of a semester. The university's manpower and other resources are heavily occupied. Thus, an organizing committee was formed, consisting of faculty teaching the computer graphic course to oversee and streamline the various activities of the exhibition. Their first task was to identify the activities and to mobilize the student force to share and execute them. For this purpose, subcommittees were formed consisting of students. However, care was taken that this digression should not cause any interruption in their studies. Students were assigned technical work like designing banners, posters, badges, monogram, and certificates so that they could become more exposed to VC environment. Some of the groups were also assigned some non-technical tasks like obtaining sponsorship and media/newspaper coverage. This was done to inform the students about the necessity of these ingredients in such events and for the publicity of the exhibition. Figure 1 and Figure 2, respectively, show the banner and the badge/exhibition emblem designed by the respective committees.

Apart from the above mentioned organizational details, the role of the exhibition is found to be very important in:

(i) Creating the business activity: Business activities carried out in this course in connection with the exhibition can be classified into two types: a) internal and b) external. External business activities were those in which the business professionals from outside the university were involved and mainly represented the expenses incurred in the exhibition. On the other hand, internal business activities were those in which the profit was generated as a result of selling landscape frames during the exhibition. The cost of frames was borne by the students themselves and was roughly around $2.32 each, including the color printout. For selling purposes and for prize distribution, picture frames were categorized into five types (1-5).

Figure 1. Exhibition Banner

Figure 2. Badge/ Exhibition Emblem

(ii) Generating the revenue: Revenue generation process involved the purchase of picture frames by the visitors on the exhibition day. Generated net revenue was calculated by subtracting the cost incurred in the exhibition from the fund generated by selling the landscape frames.

3.5 Comments

Two sets of comments are obtained - one from students about their accomplishments in VC and the other from visitors about the exhibition.

4. Lab Structure & Application of CCT

The CCT helped in quick starting the project but the concrete base was provided through the lab activities, which are discussed below in a week-wise fashion. Table 1 shows the sample lab activities and the code conversion assignments for the eight weeks along with the important information/instruction disseminated to the students. Initially, students were embarrassed by the enormity of built-in code in the skeleton VC projects. But their fears were soon dissipated as they found that they had to work only on View::OnDraw method and the view class. Students appreciated the use of ::OnDraw method as the top level of their programs. Lab exam results are not included in this research because of their non-contribution to the CCT. Sparing the code conversion assignments, exhibition announcements, and the VC environment, the format, complexity, and the schedule of lab activities was similar to the previous course conducted in 2008.

5. Results & Observations

Before going into the details of the impact of this research, it is necessary to mention that students were not subjected to any pre-test. They had no prior interaction with any OOP language including VC.

5.1 CCT Document

Figure 2 shows the performance of students with regard to the CCT document. There are 25 groups of students. For each group, both Round1 and Round2 performance is illustrated. Y-axis represents the total marks assigned to both the rounds (1 mark for each round). It is clear that the performance of majority of students improves towards the conclusion of Round2. There were some absent students in the Round1 within the groups constituting the sample.

Their groups suffer in Round1 but then improve in Round2 due to the fact that students are consistently monitored by CCT Document. It helps in creating self study habits among the students. The success of the CCT shows that the students have become conversant about transforming their TC programs into VC.

5.2 Project Performance & Exhibition Impact

All groups were able to complete their projects on time. The news about organizing the exhibition motivated other students studying in classes in which CGAM labs were being conducted using TC. A good number of these students also expressed their willingness to participate in the exhibition. Owing to their pressing demands, an extra lab

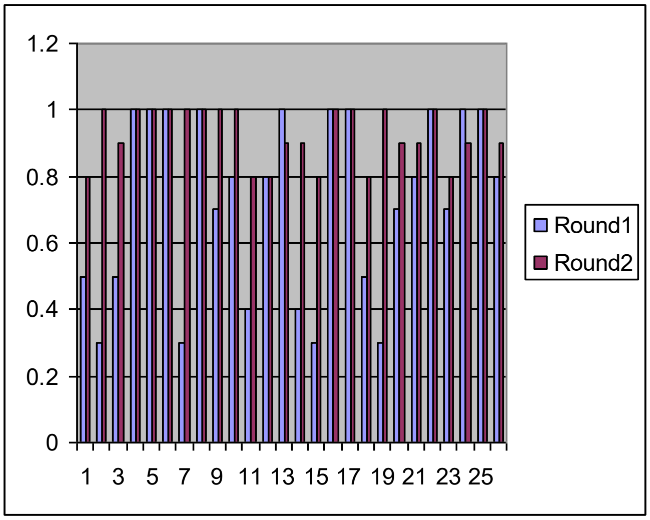

Figure 2. Graph of average Round1 and Round2 performance of all groups

session of VC was arranged for them. Twenty students attended this session but only one of the groups was able to produce a VC project.

The rest of the four groups insisted on participating and instead submitted their projects in TC. When asked why they submitted using TC they said that they needed more practice with the tool but they were interested in getting the certificates. This illustrated the fact that quick learners can benefit from CCT and the impact of supervised lab sessions cannot be denied in the learning process. Moreover, the students' interest in certificates also indicates that the exhibition strategy proved fruitful.

It is worth mentioning that initially the students were discouraged by the enormity of built-in code in the skeleton VC projects and the OOP environment. But their fears soon eased out as they found that they had to work only on View::OnDraw method and the view class. Students appreciated the use of ::OnDraw method as the top level of their program.

The project finalists are shown below. Figure 3 shows the work done by the group that won first prize, while Figure 4 and Figure 5 show the work done by runners-up in the exhibition.

5. 3 Business Activity and the Funds Generated

External activities along with business professional involved in the transaction and the costs incurred are shown below in Table 2.

Table 1. *Lab structure for eight weeks*

Week #	Topic	Important Info and Announcements	Sample Lab Activities	Code Conversion Asignment
1	Intro to Visual Studio 5 (VS5), MSDN	Path of Sample VC project	- Run the sample VC project - Write a function to draw two intersecting lines and invoke it from View ::OnDraw	Find VC counterpart of TC fuctions: line, lineto, moveto and arc using MSDN
2	Polygon Drawing and Filling	Date of Computer Artist Exhibition, committees info and assigned task	Write user-defined methods for following tasks and invoke them from View::OnDraw: - Draw an outline of a triangle using blue colored lines - Fill the triangle using green color	Find VC counterpart of TC functions: bar, bar3d, and circle using MSDN
3	2D-Translation	Group Info, Assigned Project Info, TC projects, Deadline and Requirement for project submission for Exhibition	Write user-defined methods for following tasks and invoke them from View::OnDraw: - Draw an outline of a pentagon - Calculate new coordinates of pentagon after translation using a given vector	Find VC counterpart of TC functions: ellipse, drawpoly, rectangle using MSDN
4	2D-Rotation	Request for designing posters, banners, certificates and emblem for Exhibition	- Write user-defined method to calculate the new coordinates of pentagon after fixed point rotation and invoke it from View::OnDraw.	Find VC counterpart of TC functions: cos, sin, getmaxx, and getmaxy using MSDN
5	2D- Scaling	Request for submission of final banner and emblem for Exhibition	- Write user-defined method to calculate the new coordinates of pentagon after fixed point scaling and invoke it from View::OnDraw.	Find VC counterpart of TC: putpixel, and getpixel, getcolor, setcolor using MSDN
6	Lab Exam: Translation, Scaling, Rotation	Sending of invitations	- Write a program to perform two basic transformation on an n-sided polygon	Find VC counterpart of getfillpattern, setfillpattern, getfillsettings, and setfillsettings
7	Line generation: DDA, Incremental, and Bresenham Algorithm	Official SSUET's press release for Exhibition	- Write a program to implement DDA algorithm by creating a user-defined function. Invoke the function from View::OnDaw method.	Find VC counterpart of fillellipse, fillpoly, and floodfill
8	Circle generation: Standard eq., polar form, and Bresenham Algorithm	Exhibition hall arrangement info	- Write a program to generate a circle using Bresenham Algorithm by creating a user-defined function. Invoke the function from View::OnDaw method.	Find VC counterpart of clearviewport, getimage, putimage, imagesize, memcpy

Nil cost in Table 2 indicates that the task had been completed by incorporating university professionals in order to minimize the cost of the whole operation. There were all together 32 picture frames displayed in the exhibition. Category-wise classification of the frames along with the quantity sold out and the funds so generated is shown below in Table 3.

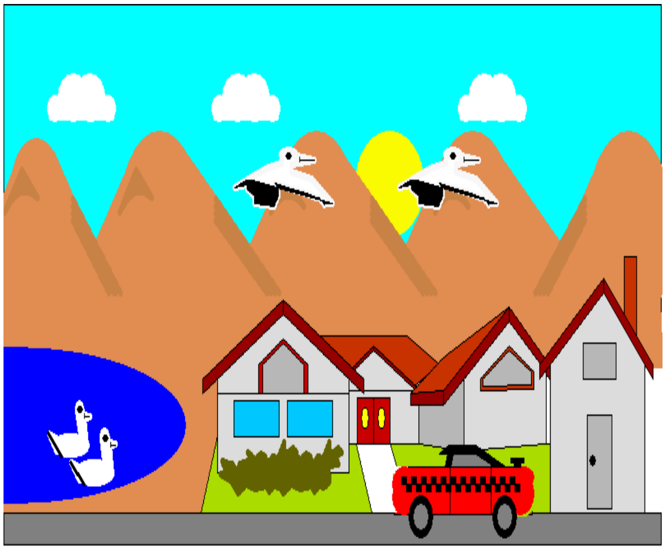

Figure 3. Output of the project obtaining first position

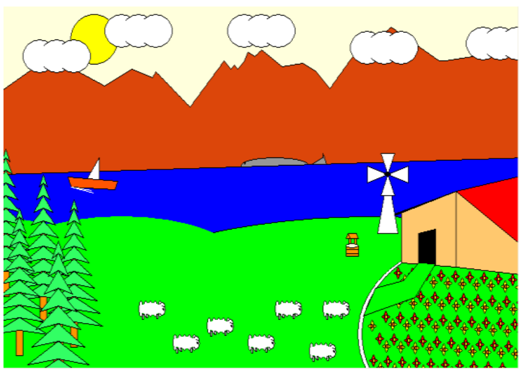

Figure 5. Second position

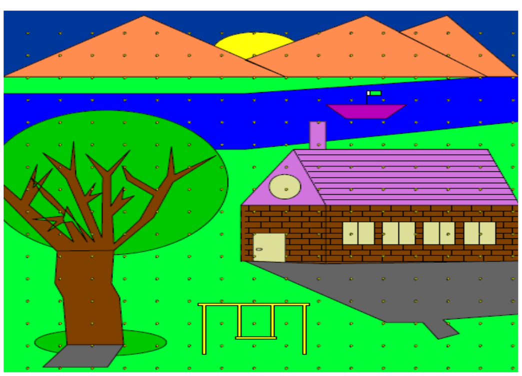

Figure 4. Third position

Thus, the total fund generated by selling the picture frames was only $54. This does not show profit as the cost for organizing the exhibition exceeded this amount. However, the objective was not to generate a profit but to give exposure to the IT students about organizing an exhibition and to participate in an economic activity.

Table 2. *Description of external business activities*

S. No.	Activity Name	Business Professional	Quantity	Costs in $
1.	Banner Printing	Printer/Publisher	3	10.46
2.	Poster Printing	Computer Operator	15	20.93
3.	Badge Printing and coating	Computer Operator and Plastic Coater	10	5.81
4.	Exhibition Hall Arrangement	Interior Decorator	Not applicable	Nil
5.	Prizes and Gifts	Salesman	32	34.88
6.	Photography (Photos)	Photographer	10	Nil

Table 3. *Description of internal business activities*

Category	Quantity (On hand)	Quantity (Sold out)	Price/Unit	Fund Generated in $
1	04	04	400	18.6
2	04	03	350	12.20
3	18	05	300	17.44
4	06	02	250	5.81

6. Comparative Study

A comparative study also has been conducted to show how the generation of business activity improved the students' overall performance in 2009 as compared to the course taught in 2008. Business activity was generated by selling VC landscape products in the exhibition. Apart from the three variables (business activity, VC tool, and the exhibition event) other parameters like the instructor, computing facility, and lab work remained the same for the two courses. It can be argued that the improvements in the course offered in 2009 are attributed to the above three variables as evident from the following facts:

i) Project Completion
Distribution of certificates in the Exhibition created an environment of interest among the students. Thus, all student groups completed their projects on time as compared to the class of 2008 in which two groups of regular students were not able to complete their projects.

ii) Imagination and Creativity
As compared to the class of 2008, students demonstrated more imagination and creativity in their work. The theme was convincingly represented in rural as well as in suburban societies. This was because students wanted to sell their products and this behavior created a competition among the students. In 2008, students focused only on the sea beaches. More engineering objects like bridges, tractors, windmills, and chimneys were included by the class of 2009, in part because that was also a requirement of the exhibition.

iii) Learning achievements
In 2008 students did not learn any new programming language but in 2009 students learned how to do graphics programming in VC in around eight weeks time period during the lab sessions using the CCT.

7. Comments

One of the comments made by a student: "These new techniques are the focus of computer art exhibition. Excellent work."

One of the comments made by the visitors of the exhibition: "Fabulous and creative work. Nice theme. It shows how much talented students are. It also shows that students can do great job in all fields."

Almost all the comments were positive and encouraging. This was because students wanted new strategies for teaching. Just delivering lectures often becomes very dry. Students must be provided opportunities to display their talents.

8. Limitation

An inherent drawback of the study was the consideration of relatively trivial content involving TC, 2-D landscapes, and a small segment of VC component. A more complex scenario based upon 3-D landscapes is being planned for future study.

9. Conclusion

Currently, world economies are bruised by the global economic downturn. Instead of going for massive projects, we can promote small businesses even through unconventional means. The teaching field also can be cultivated for this service. Teaching of science and technical courses can be inclined to breed businesses by organizing exhibitions. Mass scale adoption of this approach could decrease job losses. Along with educating students, it can provide them with pragmatic guidelines on how to start a business and how to win customer satisfaction. As far as teaching of VC is concerned, which is considered a difficult programming language, it should be taught in phases by promoting self study habits. Future work is planned to use Visual C for creating 3-D landscapes.

10. Acknowledgement

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